Technical Assignment: Advisor Booking System

Objective

You have recently joined a team as a Senior Java Developer at a financial advisory company. The business has identified an opportunity to improve the customer experience by making it easier to book meetings with advisors.

Today, appointments are coordinated manually through phone calls and email exchanges, creating unnecessary delays for both customers and advisors. To address this, the company wants to introduce a centralized booking capability that allows customers to reserve available timeslots directly from existing digital channels.

The company already has a Microsoft 365 ecosystem in place and relies on Outlook and Exchange Online for advisor calendars and email communication. Rather than replacing existing systems, the business wants a solution that integrates seamlessly with the current environment while providing a reliable and scalable booking experience.

Customers must be able to book meetings through:

1. An existing public website
2. An existing mobile application

The booking solution must be capable of:

1. Receiving a booking request from either channel.
2. Checking the advisor's calendar availability.
3. Creating an appointment if the selected slot is available.
4. Storing the booking in a database.
5. Preventing double-booking when multiple customers attempt to book the same timeslot simultaneously.
6. Sending a confirmation email containing meeting details, including date, time, and location.

The business has asked you to propose a high-level technical solution and architecture that satisfies these requirements while ensuring reliability, maintainability, and future scalability. You are free to choose technologies and architectural patterns that you believe are appropriate, but your recommendations should be justified based on the business context and requirements provided.

Because of past bad experiences, the business would also has explicitly asked you to describe step by step what happens inside the booking service from the moment a request arrives until the booking is confirmed. Whether you choose to show this in a flow, sequence, or similar diagram is up to you. It just needs to be documented.

Technical Assignment: Advisor Booking System

Your Task

Create a high-level solution architecture that satisfies the business requirements.

You may choose any technologies you consider appropriate and assume reasonable conditions where information is missing but make sure to note them.

Write a 1-2-page document that describes your solution and answers the questions throughout this assignment. This does not have to be heavy reading – notes that reference parts of the diagram or directly answer the questions, is perfectly fine.

A simple sketch or whiteboard-style diagram is sufficient. Diagram aesthetics will not be evaluated, though it must be clearly labelled and reader friendly. You are of course welcome to use any digital tools at your disposal, if you see those more fit.

No implementation is required.

1. Solution Overview and Implementation Approach

Provide a brief description of your proposed solution.

Note down in your 1-2 pager:

- Main architectural style (e.g., monolith, microservices, serverless)
- Key technologies selected
- If you were implementing this in Java, which major components/classes/services would you expect to build?

We will then discuss on the next interview, why you made those choices.

2. Architecture Diagram

Create a simple architecture diagram showing how the:

- Website
- Mobile App
- Backend/API
- Exchange/Calendar integration
- Database
- Email service
- Any additional components you consider necessary

Technical Assignment: Advisor Booking System

3. Handling Race Conditions

One of the most important requirements is preventing double-booking.
Consider the following scenario:

- Customer A books Advisor X at 10:00.
- Customer B books the same advisor at 10:00 milliseconds later.

Note down in your 1-2 pager:

- How your solution prevents duplicate bookings.
- What mechanisms ensure consistency.
- How users are informed when a slot becomes unavailable.

4. Error Handling

Describe how your solution would handle:

Scenario A - The calendar service is unavailable.

Scenario B - The database is unavailable.

Scenario C - The confirmation email fails to send.

Note down in your 1-2 pager:

- What the customer experiences.
- What is logged.
- Whether retries should occur.

5. Scalability and Future Growth

The company expects usage to grow significantly, and the system must be able to support 500,000 bookings per month across 2,000 advisors.
Consider how your solution could support:

- Booking volume
- More advisors
- Additional customer channels
- How would you monitor this solution in a production setting?

Note this down in your 1-2 pager